Tests of General Relativity with gravitational wave observations

Shilpa Kastha



From gravitational wave observations to tests of GR

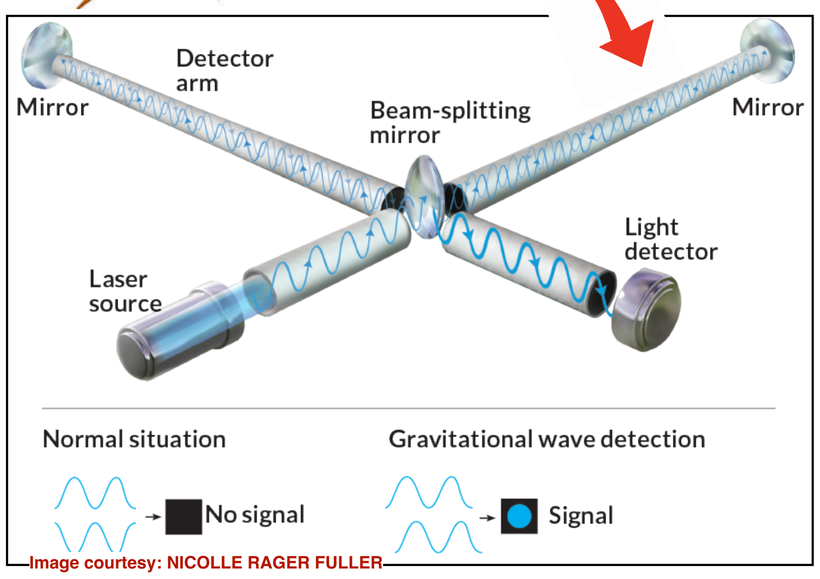
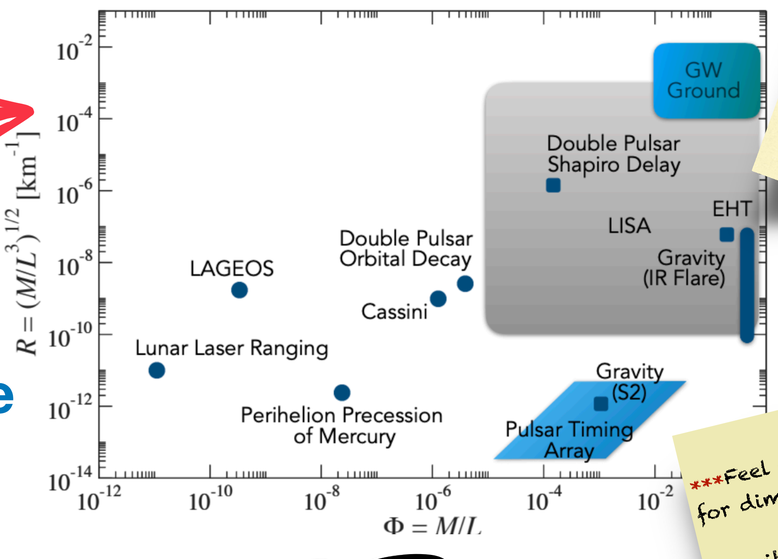


Image courtesy: NICOLLE RAGER FULLER

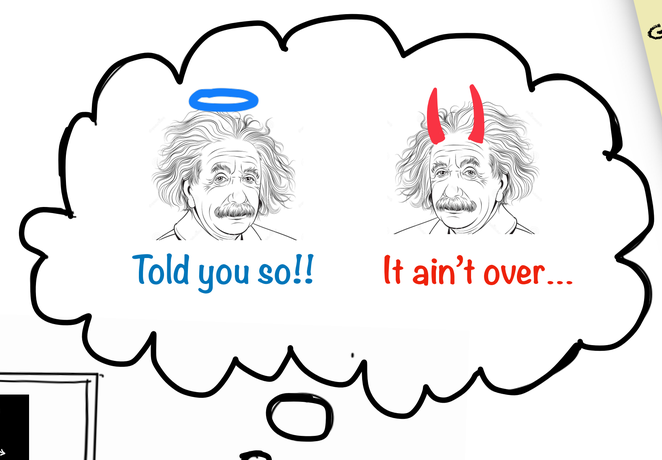
arXiv:1903.09221v3

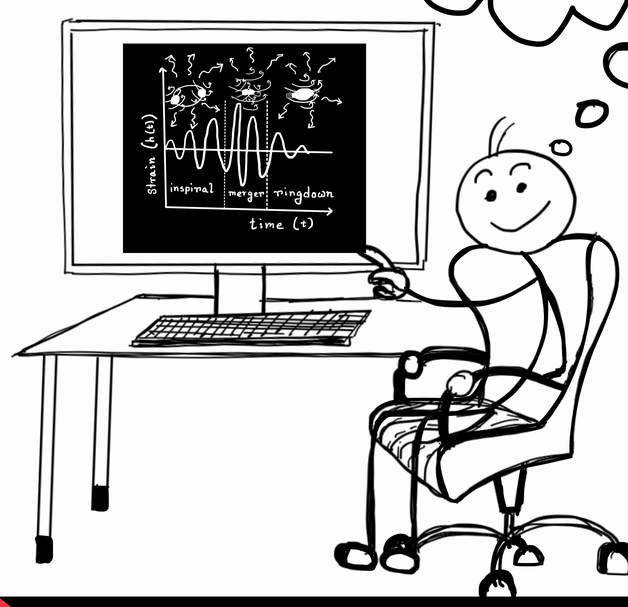


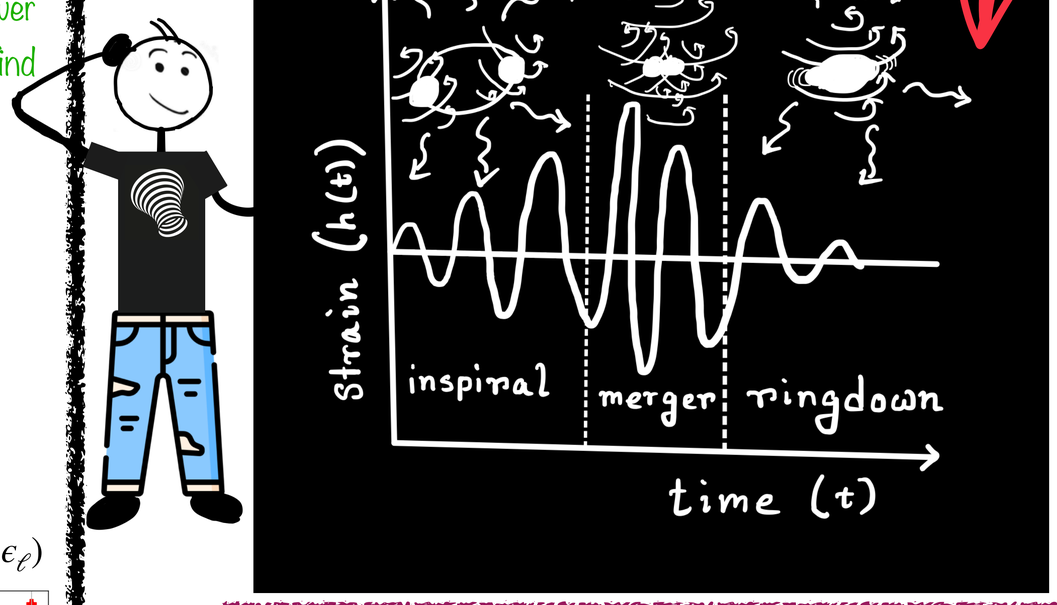
For any queries / comments
email : shilpa.kaslka@nbi.ku.dk

***Feel free to incorporate G and c for dimensional consistency..

Gravitational wave jargon
- GW : Gravitational waves
- GR: General Relativity
- SPA : Stationery phase approximation
- BH : Black hole
- PN : Post-Newtonian
- IMR : Inspiral-merger-ringdown
- PE : Parameter estimation
- LIGO : Laser Interferometer Gravitational wave Observatory







Inspiral tests

PN structure of the parametrised phasing formula in Frequency domain using SPA

$$\tilde{h}(f) = \mathcal{A} f^{-7/6} e^{i\psi_f(f)}$$

$$\psi_f(t_f) = 2\pi f t_c - \phi_c + \frac{3}{128\nu v^5}\left[\sum_{k=0}^{N} \psi_k v^{k-5}\right]$$

Before looking for a first principle approach to newer theories, the immediate but obvious choice is to find plausible deviations from GR.

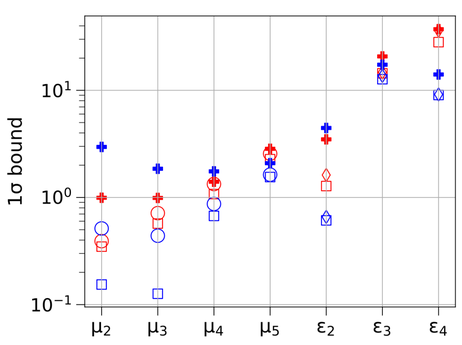
Introducing deviation at the level of multipole moment

$$M_{\langle ij\rangle} = \mu_2 M^{\mathcal{GR}}_{\langle ij\rangle}$$
$$M_{\langle ijk\rangle} = \mu_3 M^{\mathcal{GR}}_{\langle ijk\rangle}$$
$$M_{\langle ijkl\rangle} = \mu_4 M^{\mathcal{GR}}_{\langle ijkl\rangle}$$
$$M_{\langle ijklm\rangle} = \mu_5 M^{\mathcal{GR}}_{\langle ijklm\rangle}$$

$$S_{\langle ij\rangle} = \epsilon_2 S^{\mathcal{GR}}_{\langle ij\rangle}$$
$$S_{\langle ijk\rangle} = \epsilon_3 S^{\mathcal{GR}}_{\langle ijk\rangle}$$
$$S_{\langle ijkl\rangle} = \epsilon_4 S^{\mathcal{GR}}_{\langle ijkl\rangle}$$

$\mathcal{A}(\nu, f, m) \to \mathcal{A}(\nu, f, m, \mu_\ell, \epsilon_\ell)$ and $\psi_k(\nu) \to \psi_k(\nu, \mu_\ell, \epsilon_\ell)$

projected 1-σ bounds on one of the non-GR parameters when other parameters are kept at their GR values (= 1) for a network of detectors consists of LIGO-Hanford, LIGO-Livingston and Virgo



Multiband observations with LISA and CE

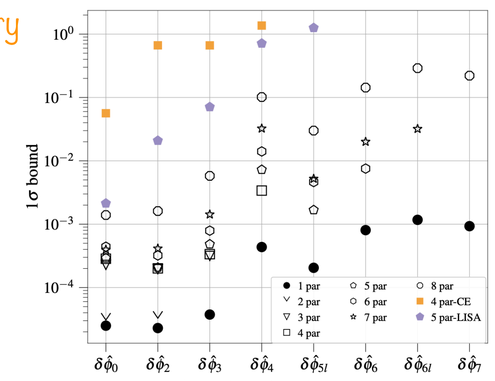
Parametrised tests for PN theory

Deviations are added to the PN coefficient

$$\psi_k \to \psi_k(1 + \delta\hat{\phi}_k)$$

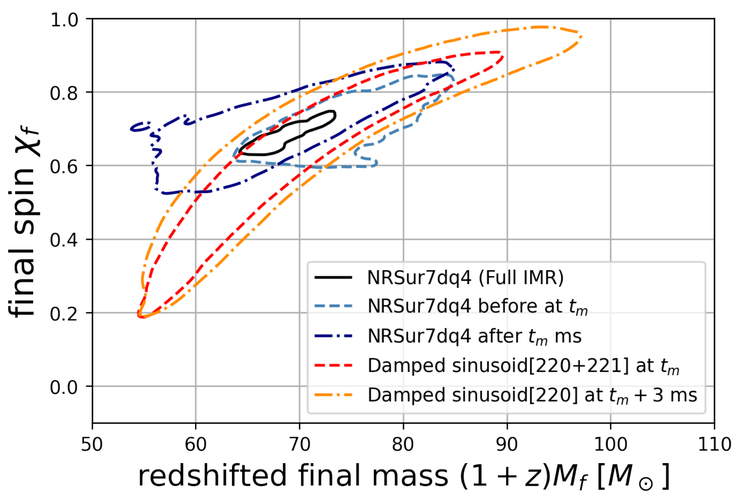
Non GR Parameters

Shown are combined 1σ bounds on various PN coefficients while measuring many of them together for a BBH population having Gaussian spin distribution.



IMR consistency tests

PE from the inspire and the post merger data



Tests Hawking's BH area theorem

$$A_1 + A_2 \equiv A_i < A_f.$$

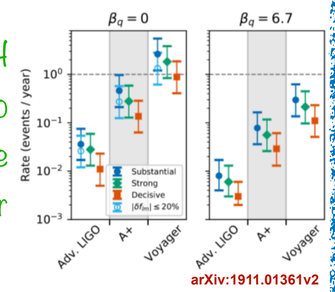
Ringdown test

$$h(t) = \frac{M}{D_L}\sum_{\ell mn} {}_{-2}S_{\ell m}(\iota, \varphi)\, A_{\ell mn}\, e^{i(\Omega_{\ell mn} t + \phi_{\ell mn})}$$

$$\Omega_{\ell mn} = 2\pi f_{\ell mn} + i/\tau_{\ell mn}$$

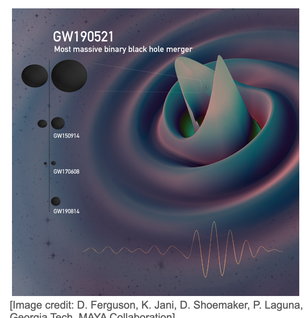
No-hair theorem : $\Omega_{\ell mn} = \Omega_{\ell mn}(M, \chi)$

Expected rates of BBH mergers for which two ringdown modes can be observed and resolved For two population model

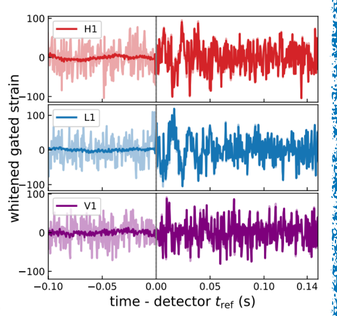


arXiv:1911.01361v2

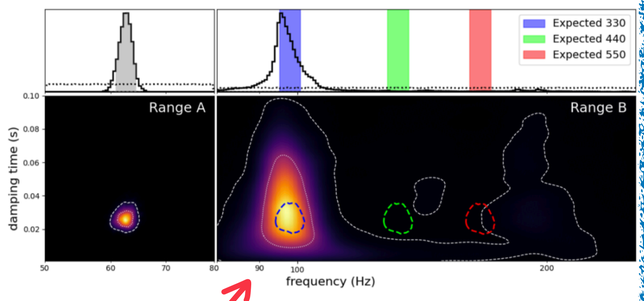
GW190521



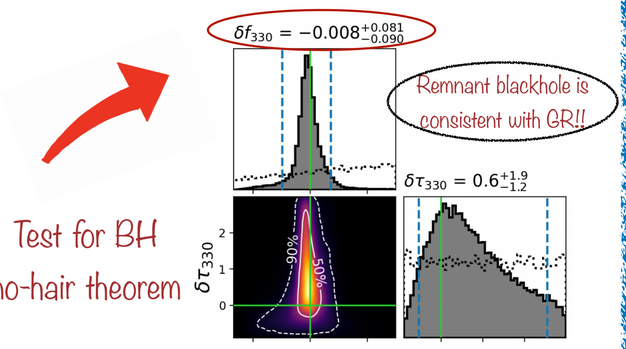
[Image credit: D. Ferguson, K. Jani, D. Shoemaker, P. Laguna, Georgia Tech, MAYA Collaboration]



BH spectroscopy



Secondary mode found



Test for BH no-hair theorem

We find good agreement with GR (so far !!!)

strong. niels bohr institute
KØBENHAVNS UNIVERSITET
Danmarks Grundforskningsfond Danish National Research Foundation
VILLUM FONDEN
erc European Research Council